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Exno:1

Data Cleaning Process

AIM

To read the given data and perform data cleaning and save the cleaned data to a file.

Explanation

Data cleaning is the process of preparing data for analysis by removing or modifying data that is incorrect ,incompleted , irrelevant , duplicated or improperly formatted. Data cleaning is not simply about erasing data ,but rather finding a way to maximize datasets accuracy without necessarily deleting the information.

Algorithm

STEP 1: Read the given Data

STEP 2: Get the information about the data

STEP 3: Remove the null values from the data

STEP 4: Save the Clean data to the file

STEP 5: Remove outliers using IQR

STEP 6: Use zscore of to remove outliers

Coding and Output

from google.colab import drive

drive.mount("/content/drive")

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

ls drive/MyDrive/'Colab Notebooks'/Data_set.csv

```
'drive/MyDrive/Colab Notebooks/Data_set.csv'
```

import pandas as pd

df=pd.read_csv('drive/MyDrive/Colab Notebooks/Data_set.csv')

df

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
0	NaN	South Korea	16	Friday, Saturday	tvN	8.9	33.0	1	111706.0
1	NaN	South Korea	16	Friday, Saturday	jTBC	8.7	89.0	2	100950.0
2	Descendants of the Sun	South Korea	16	Wednesday, Thursday	KBS2	8.7	77.0	3	96318.0
3	Boys Over Flowers	South Korea	25	Monday, Tuesday	KBS2	7.7	2249.0	4	94228.0
4	W	South Korea	16	Wednesday, Thursday	MBC	8.5	201.0	5	92121.0
...	...	...	...	...	...	...	...	...	...
95	Shut Up: Flower Boy Band	South Korea	16	Monday, Tuesday	tvN	8.1	806.0	99	34668.0
96	Blood	South Korea	20	Monday, Tuesday	KBS2	7.4	3271.0	100	34666.0
97	Chicago Typewriter	South Korea	16	Friday, Saturday	tvN	8.8	51.0	101	NaN
98	Sungkyunkwan Scandal	South Korea	20	Monday, Tuesday	KBS2	8.2	605.0	102	34615.0
99	Vagabond	South Korea	16	Friday, Saturday	SBS, Netflix	8.5	238.0	103	34523.0

100 rows × 9 columns

df.isnull().sum()

	0
show_name	4
country	0
num_episodes	0
aired_on	1
original_network	1
rating	4
current_overall_rank	3
lifetime_popularity_rank	0
watchers	3

dtype: int64

df.isnull().dropna()

Code cell output actions

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
		False	False	False	False	False	False	False	False
1	True	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...
95	False	False	False	False	False	False	False	False	False
96	False	False	False	False	False	False	False	False	False
97	False	False	False	False	False	False	False	False	True
98	False	False	False	False	False	False	False	False	False
99	False	False	False	False	False	False	False	False	False

100 rows × 9 columns

df.fillna(0)

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
0	0	South Korea	16	Friday, Saturday	tvN	8.9	33.0	1	111706.0
1	0	South Korea	16	Friday, Saturday	jTBC	8.7	89.0	2	100950.0
2	Descendants of the Sun	South Korea	16	Wednesday, Thursday	KBS2	8.7	77.0	3	96318.0
3	Boys Over Flowers	South Korea	25	Monday, Tuesday	KBS2	7.7	2249.0	4	94228.0
4	W	South Korea	16	Wednesday, Thursday	MBC	8.5	201.0	5	92121.0
...	...	...	...	...	...	...	...	...	...
95	Shut Up: Flower Boy Band	South Korea	16	Monday, Tuesday	tvN	8.1	806.0	99	34668.0
96	Blood	South Korea	20	Monday, Tuesday	KBS2	7.4	3271.0	100	34666.0
97	Chicago Typewriter	South Korea	16	Friday, Saturday	tvN	8.8	51.0	101	0.0
98	Sungkyunkwan Scandal	South Korea	20	Monday, Tuesday	KBS2	8.2	605.0	102	34615.0
99	Vagabond	South Korea	16	Friday, Saturday	SBS, Netflix	8.5	238.0	103	34523.0

100 rows × 9 columns

df.ffill()

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
0	NaN	South Korea	16	Friday, Saturday	tvN	8.9	33.0	1	111706.0
1	NaN	South Korea	16	Friday, Saturday	jTBC	8.7	89.0	2	100950.0
2	Descendants of the Sun	South Korea	16	Wednesday, Thursday	KBS2	8.7	77.0	3	96318.0
3	Boys Over Flowers	South Korea	25	Monday, Tuesday	KBS2	7.7	2249.0	4	94228.0
4	W	South Korea	16	Wednesday, Thursday	MBC	8.5	201.0	5	92121.0
...	...	...	...	...	...	...	...	...	...
95	Shut Up: Flower Boy Band	South Korea	16	Monday, Tuesday	tvN	8.1	806.0	99	34668.0
96	Blood	South Korea	20	Monday, Tuesday	KBS2	7.4	3271.0	100	34666.0
97	Chicago Typewriter	South Korea	16	Friday, Saturday	tvN	8.8	51.0	101	34666.0
98	Sungkyunkwan Scandal	South Korea	20	Monday, Tuesday	KBS2	8.2	605.0	102	34615.0
99	Vagabond	South Korea	16	Friday, Saturday	SBS, Netflix	8.5	238.0	103	34523.0

100 rows × 9 columns

df.bfill()

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
0	Descendants of the Sun	South Korea	16	Friday, Saturday	tvN	8.9	33.0	1	111706.0
1	Descendants of the Sun	South Korea	16	Friday, Saturday	jTBC	8.7	89.0	2	100950.0
2	Descendants of the Sun	South Korea	16	Wednesday, Thursday	KBS2	8.7	77.0	3	96318.0
3	Boys Over Flowers	South Korea	25	Monday, Tuesday	KBS2	7.7	2249.0	4	94228.0
4	W	South Korea	16	Wednesday, Thursday	MBC	8.5	201.0	5	92121.0
...	...	...	...	...	...	...	...	...	...
95	Shut Up: Flower Boy Band	South Korea	16	Monday, Tuesday	tvN	8.1	806.0	99	34668.0
96	Blood	South Korea	20	Monday, Tuesday	KBS2	7.4	3271.0	100	34666.0
97	Chicago Typewriter	South Korea	16	Friday, Saturday	tvN	8.8	51.0	101	34615.0
98	Sungkyunkwan Scandal	South Korea	20	Monday, Tuesday	KBS2	8.2	605.0	102	34615.0
99	Vagabond	South Korea	16	Friday, Saturday	SBS, Netflix	8.5	238.0	103	34523.0

100 rows × 9 columns

df_mean1=df['num_episodes'].fillna(df['num_episodes'].mean())

df_mean1

	num_episodes
0	16
1	16
2	16
3	25
4	16
...	...
95	16
96	20
97	16
98	20
99	16

100 rows × 1 columns

dtype: int64

df_mean2=df['rating'].fillna(df['rating'].mean())

df_mean2

	rating
0	8.9
1	8.7
2	8.7
3	7.7
4	8.5
...	...
95	8.1
96	7.4
97	8.8
98	8.2
99	8.5

100 rows × 1 columns

dtype: float64

df_mean3=df['current_overall_rank'].fillna(df['current_overall_rank'].mean())

df_mean3

	current_overall_rank
0	33.0
1	89.0
2	77.0
3	2249.0
4	201.0
...	...
95	806.0
96	3271.0
97	51.0
98	605.0
99	238.0

100 rows × 1 columns

dtype: float64

df_mean4=df['lifetime_popularity_rank'].fillna(df['lifetime_popularity_rank'].mean())

df_mean4

	lifetime_popularity_rank
0	1
1	2
2	3
3	4
4	5
...	...
95	99
96	100
97	101
98	102
99	103

100 rows × 1 columns

dtype: int64

df_mean5=df['watchers'].fillna(df['watchers'].mean())

df_mean5

	watchers
0	111706.000000
1	100950.000000
2	96318.000000
3	94228.000000
4	92121.000000
...	...
95	34668.000000
96	34666.000000
97	52994.907216
98	34615.000000
99	34523.000000

100 rows × 1 columns

dtype: float64

df_dropna=df.dropna()

df_dropna

	show_name	country	num_episodes	aired_on	original_network	rating	current_overall_rank	lifetime_popularity_rank	watchers
2	Descendants of the Sun	South Korea	16	Wednesday, Thursday	KBS2	8.7	77.0	3	96318.0
3	Boys Over Flowers	South Korea	25	Monday, Tuesday	KBS2	7.7	2249.0	4	94228.0
4	W	South Korea	16	Wednesday, Thursday	MBC	8.5	201.0	5	92121.0
5	You Who Came from the Stars	South Korea	21	Wednesday, Thursday	SBS	8.6	112.0	6	91360.0
6	Weightlifting Fairy Kim Bok Joo	South Korea	16	Wednesday, Thursday	MBC	8.8	40.0	7	91330.0
...	...	...	...	...	...	...	...	...	...
94	Flower of Evil	South Korea	16	Wednesday, Thursday	tvN	9.1	4.0	98	34901.0
95	Shut Up: Flower Boy Band	South Korea	16	Monday, Tuesday	tvN	8.1	806.0	99	34668.0
96	Blood	South Korea	20	Monday, Tuesday	KBS2	7.4	3271.0	100	34666.0
98	Sungkyunkwan Scandal	South Korea	20	Monday, Tuesday	KBS2	8.2	605.0	102	34615.0
99	Vagabond	South Korea	16	Friday, Saturday	SBS, Netflix	8.5	238.0	103	34523.0

86 rows × 9 columns

import pandas as pd

import seaborn as sns

age=[1,3,28,27,25,92,30,39,40,50,26,24,29,94]

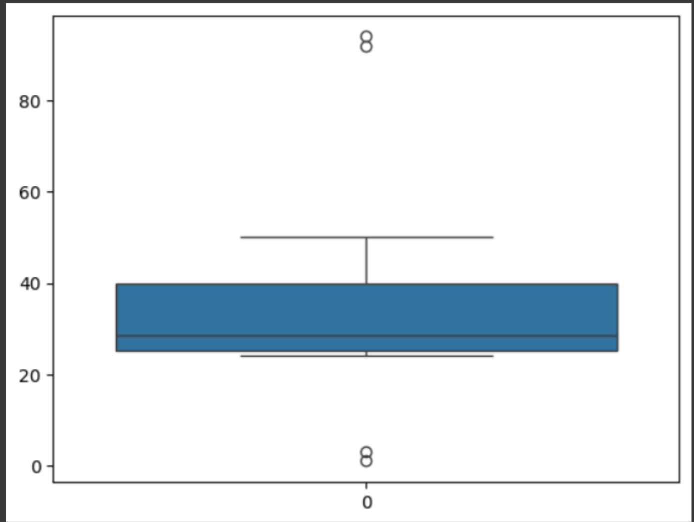
af=pd.DataFrame(age)

af

	0
0	1
1	3
2	28
3	27
4	25
5	92
6	30
7	39
8	40
9	50
10	26
11	24
12	29
13	94

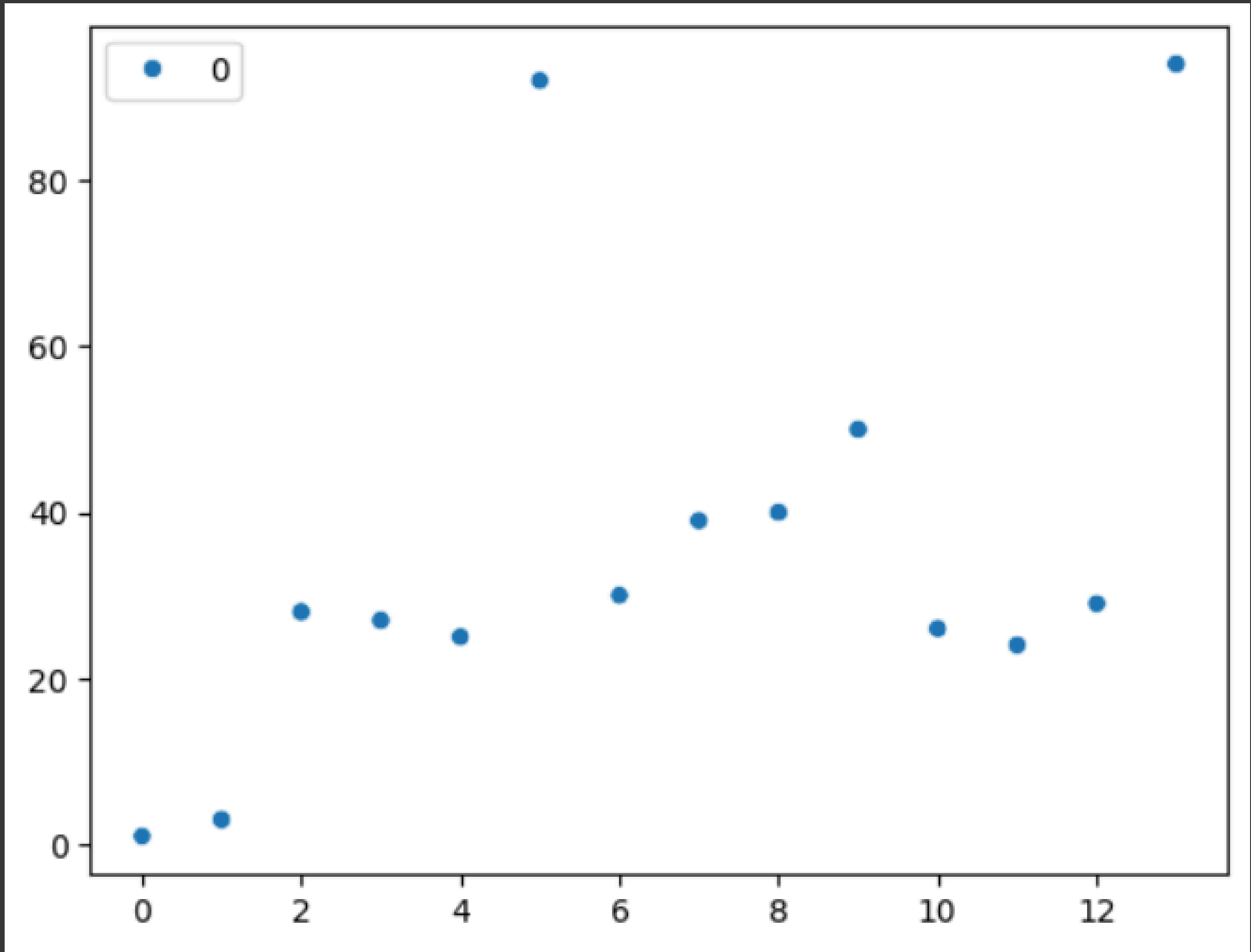
sns.boxplot(af)

```
/usr/local/lib/python3.10/dist-packages/seaborn/_base.py:949: FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future v
  data_subset = grouped_data.get_group(pd_key)
/usr/local/lib/python3.10/dist-packages/seaborn/categorical.py:640: FutureWarning: SeriesGroupBy.grouper is deprecated and will be removed in a future version of pandas.
  positions = grouped.grouper.result_index.to_numpy(dtype=float)
<Axes: >
```



sns.scatterplot(af)

<Axes: >



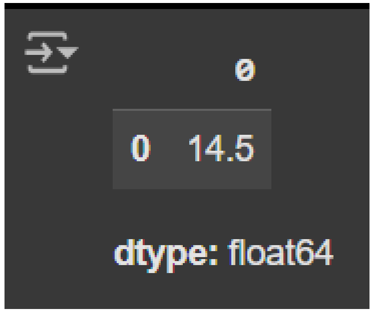
q1=af.quantile(0.25)

q2=af.quantile(0.5)

q3=af.quantile(0.75)

iqr=q3-q1

iqr



import numpy as np

Q1=np.percentile(af,25)

Q2=np.percentile(af,50)

Q3=np.percentile(af,75)

IQR=Q3-Q1

lower_bound=Q1-1.5*IQR

upper_bound=Q3+1.5*IQR

outliers = [x for x in age if x < lower_bound or x > upper_bound]

print('Q1:',Q1)

print('Q3:',Q3)

print('IQR:',IQR)

print('Lower bound:',lower_bound)

print('Upper bound:',upper_bound)

print('Outliers:',outliers)

```
Q1: 25.25
Q3: 39.75
IQR: 14.5
Lower bound: 3.5
Upper bound: 61.5
Outliers: [1, 3, 92, 94]
```

af=af[((af>=lower_bound)&(af<=upper_bound))]

af

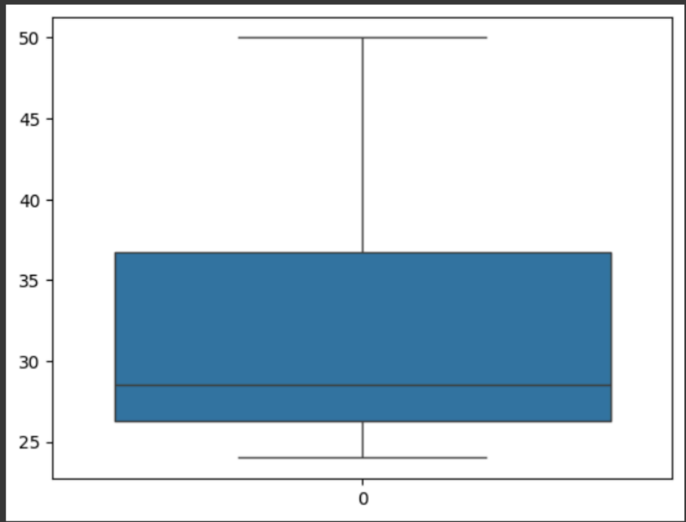
	0
0	NaN
1	NaN
2	28.0
3	27.0
4	25.0
5	NaN
6	30.0
7	39.0
8	40.0
9	50.0
10	26.0
11	24.0
12	29.0
13	NaN

af.dropna()

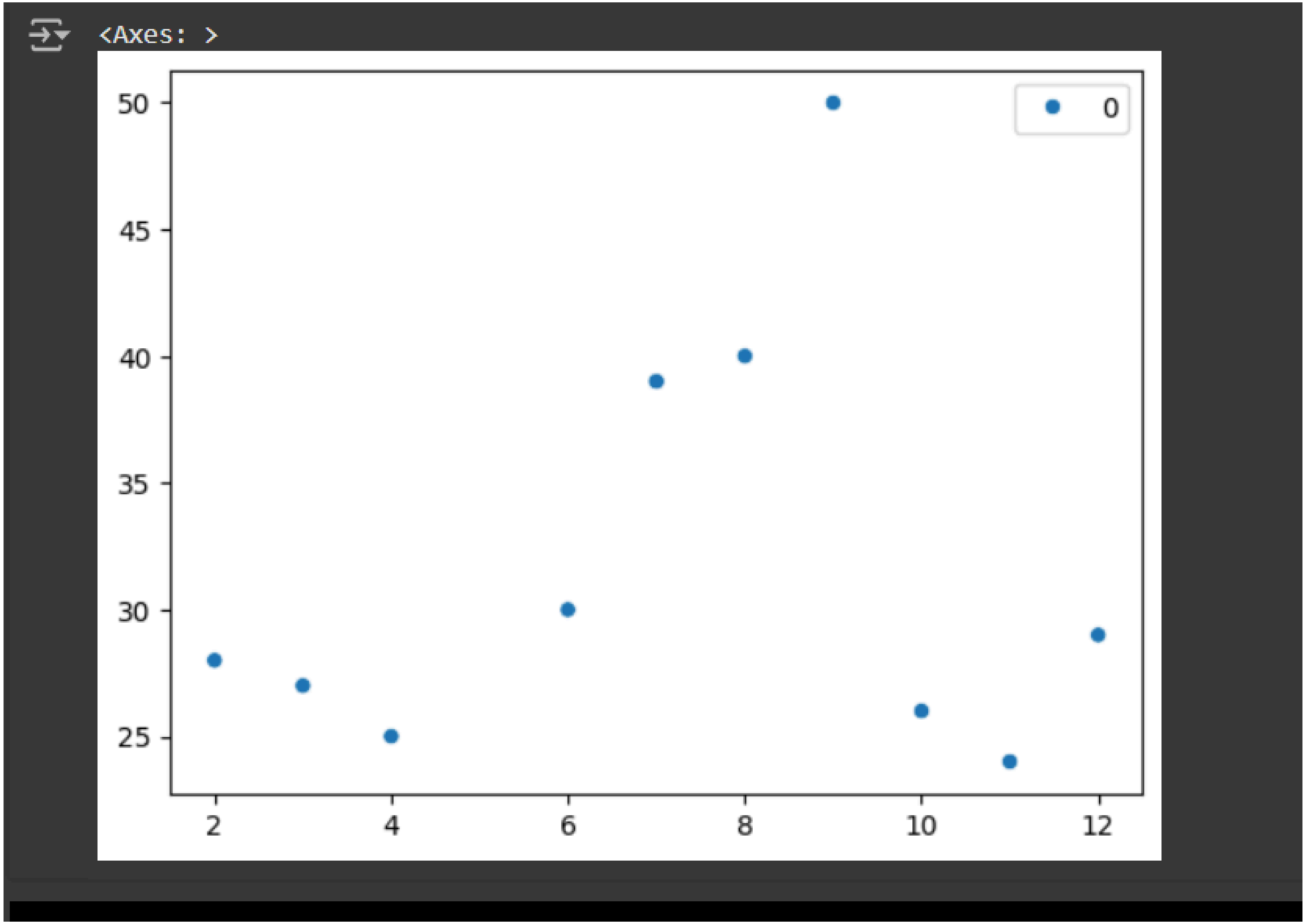
	0
2	28.0
3	27.0
4	25.0
6	30.0
7	39.0
8	40.0
9	50.0
10	26.0
11	24.0
12	29.0

sns.boxplot(af)

```
/usr/local/lib/python3.10/dist-packages/seaborn/_base.py:949: FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future ve
  data_subset = grouped_data.get_group(pd_key)
/usr/local/lib/python3.10/dist-packages/seaborn/categorical.py:640: FutureWarning: SeriesGroupBy.grouper is deprecated and will be removed in a future version of pandas.
  positions = grouped.grouper.result_index.to_numpy(dtype=float)
<Axes: >
```



sns.scatterplot(af)



from scipy import stats

import numpy as np

import pandas as pd

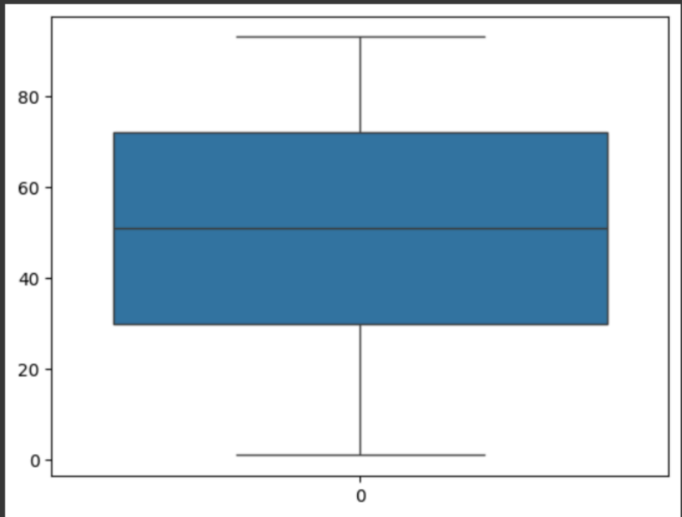
import seaborn as sns

data=[1,12,15,18,21,24,27,30,33,36,39,42,45,48,51,54,57,60,63,66,69,72,75,78,81,84,87,90,93]

df=pd.DataFrame(data)

sns.boxplot(df)

```
/usr/local/lib/python3.10/dist-packages/seaborn/_base.py:949: FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future ve
  data_subset = grouped_data.get_group(pd_key)
/usr/local/lib/python3.10/dist-packages/seaborn/categorical.py:640: FutureWarning: SeriesGroupBy.grouper is deprecated and will be removed in a future version of pandas.
  positions = grouped.grouper.result_index.to_numpy(dtype=float)
<Axes: >
```



mean=np.mean(data)

mean

std=np.std(data)

std

z=np.abs(stats.zscore(df))

z

	0
0	1.942433
1	1.512727
2	1.395535
3	1.278342
4	1.161149
5	1.043957
6	0.926764
7	0.809572
8	0.692379
9	0.575187
10	0.457994
11	0.340801
12	0.223609
13	0.106416
14	0.010776
15	0.127969

threshold=3

outliers = df[abs(df) > 3]

print("Outliers:")

print(outliers)

```
       0
0    NaN
1   12.0
2   15.0
3   18.0
4   21.0
5   24.0
6   27.0
7   30.0
8   33.0
9   36.0
10  39.0
11  42.0
12  45.0
13  48.0
14  51.0
15  54.0
16  57.0
17  60.0
18  63.0
19  66.0
20  69.0
21  72.0
22  75.0
23  78.0
24  81.0
25  84.0
26  87.0
27  90.0
28  93.0
```

df_cleaned = df[(z <= threshold)]

df_cleaned

	0
0	1
1	12
2	15
3	18
4	21
5	24
6	27
7	30
8	33
9	36
10	39
11	42
12	45
13	48
14	51
15	54
16	57

sns.boxplot(df_cleaned)

```
/usr/local/lib/python3.10/dist-packages/seaborn/_base.py:949: FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future ve
  data_subset = grouped_data.get_group(pd_key)
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  positions = grouped.grouper.result_index.to_numpy(dtype=float)
<Axes: >
```
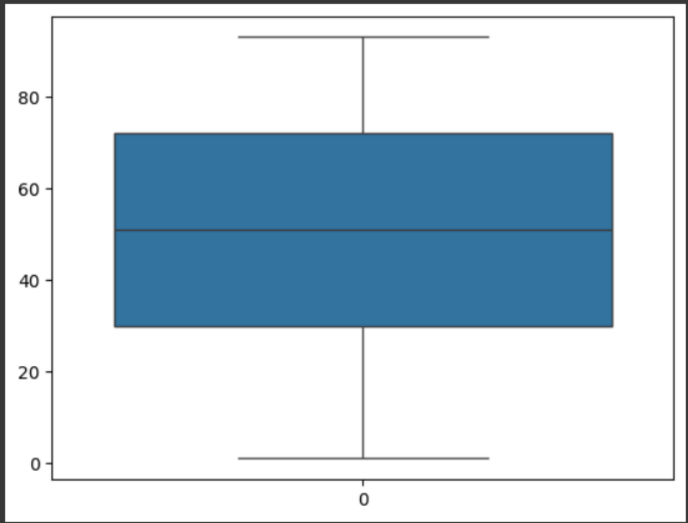
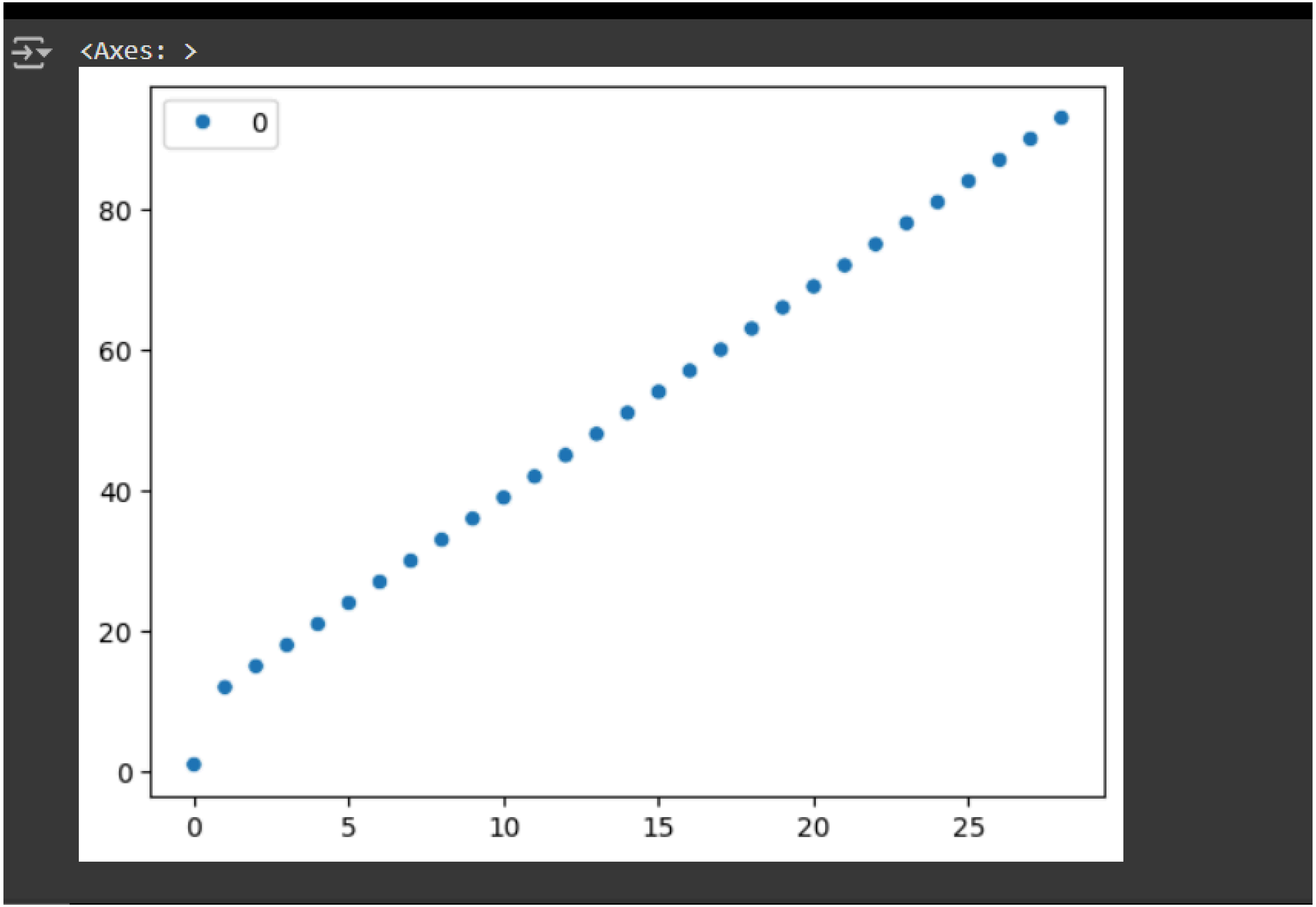



sns.scatterplot(df_cleaned)



Result

```
Thus the above code is execueted successfully...
```